AKSHITH ULLAL, Ph.D.

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EDUCATION

Vanderbilt University, Nashville, TN **2018-2024**
Ph.D. in Electrical and Computer Engineering
- Dissertation: Generation of Naturalistic Interactions for Large-Scale Avatar Telepresence in Mixed Reality
- Advisor: Nilanjan Sarkar, Ph.D.

University of Missouri, Columbia, MO **2015-2018**
M.S. in Electrical and Computer Engineering
- Thesis: Non-Invasive Monitoring of Vital Signs using Recliner Chairs; Advisor: Marjorie Skubic, Ph.D.

Visvesvaraya Technological University, Belagavi, India **2011-2015**
B.Tech. in Electronics and Communication Engineering

RESEARCH FOCUS

Develop human-centered and trustworthy artificial intelligence systems for healthcare, including multimodal clinical AI, intelligent embodied agents, digital twins, and augmented/virtual reality interfaces, with an emphasis on clinical decision support, patient communication, safety, accessibility, and real-world deployment.

PROFESSIONAL EXPERIENCE

Intelligent Clinical Care Center (IC3), University of Florida, Gainesville, FL **Jun 2024-Present**
Applications Developer Analyst V
- Develop AI-enabled hospital digital twins with LLM-powered embodied assistants, retrieval-augmented generation, speech/action interfaces, and cloud streaming for tablet and immersive deployment.
- Developed an ICU digital-twin platform with an interactive MetaHuman assistant and remote iPad access; the platform has been evaluated with patients, medical trainees, and clinical providers.
- Develop multimodal Transformer-based models integrating structured perioperative electronic health record data and clinical-note representations for prediction of postoperative complications across UF Health Gainesville and Jacksonville.
- Contribute to real-time perioperative AI infrastructure that integrates clinical data streams, model inference, and deployment monitoring for surgical risk assessment.
- Develop natural-language-controlled embodied agents capable of generating actions in AR/VR environments from user speech and contextual information.

Department of Electrical and Computer Engineering, Vanderbilt University, Nashville, TN **Oct 2024-Oct 2025**
Adjoint Assistant Professor
- Co-developed the protocol and technical workflow for a 2-arm pilot randomized controlled trial comparing collaborative augmented reality with 2D communication for older adult-companion pairs.
- Designed and refined collaborative AR activities for older adults through participatory design, producing deployable activities for later clinical evaluation.

Robotics and Autonomous Systems Lab, Vanderbilt University, Nashville, TN **Aug 2018-May 2024**
Graduate Research Assistant
- Developed multi-objective optimization approaches for redirecting avatar interactions while preserving human body pose and nonverbal behavior across dissimilar remote AR/MR environments.
- Developed mesh-deformation methods for mapping user locomotion trajectories between remote environments with irregular geometries.
- Developed avatar telepresence systems using Unreal/Unity, networking, cloud services, and photorealistic avatars; conducted human-subject evaluations of naturalistic remote interaction.
- Developed machine-learning models for sleep-stage classification in children with Rett syndrome and multimodal stress detection for older adults.

Center to Stream Healthcare in Place, University of Missouri, Columbia, MO **Aug 2016-May 2018**
Graduate Research Assistant
- Developed a non-wearable recliner-chair sensing system to estimate heart and respiratory rates from ballistocardiogram signals using deep regression methods; evaluated with 45 older adults.

Biomedical Signal Processing Lab, IIT Indore, India **May 2017-Aug 2017**
Summer Research Intern
- Developed a deep neural network for seizure classification using Variational Mode Decomposition-derived EEG features; invited to contribute a related book chapter.

PATENTS AND INVENTION DISCLOSURES

P1 U.S. Patent 12,462,501, "System and Method for Efficient Redirection of User Interactions and Gestures between Remote and Local Environments in Augmented Reality."

P2 U.S. Patent Application 19/377,510, "System and Method for Efficient Redirection of User Interactions and Gestures between Remote and Local Environments in Augmented Reality" (2026).

P3 U.S. Patent Application 18/522,027, "System and Method for Efficient Mapping of User Locomotion between Remote and Local Environments in Augmented Reality" (pending).

P4 Personalized Health and Intelligence Learning (PHIL), University of Florida invention disclosure UFTINV-260020 (patent application in progress).

P5 "System and method to generate naturalistic locomotion paths for avatar telepresence interactions that involve large size differences between user and avatar environments," Vanderbilt disclosure VU24205 (patent application in progress).

P6 "System and method to generate naturalistic walking gait body poses based on given input velocity and body joint constraints for performing naturalistic avatar-based telepresence interactions in mixed reality," Vanderbilt disclosure VU24206 (patent application in progress).

P7 U.S. Provisional Patent 62/481,968, "Non-wearable Heart Rate, Respiratory Rate and Motion Monitoring System."

PUBLICATIONS

Full publication record: Google Scholar

Peer-Reviewed Journal Articles

J1 Patel, R., Haziroglu, C., Ullal, A., Soliman, A., Seshadri, S., Efron, E., Sela, O., et al. "Artificial intelligence across the medical education continuum: institutional, learning, readiness and safety systems." npj Digital Medicine, 2026. doi:10.1038/s41746-026-03362-2.

J2 Sarkar, N., Tate, J. A., Tauseef, M., Ullal, A., Maxwell, C. A., Juckett, L. A., et al. "Feasibility and Acceptability of Collaborative Augmented Reality for Older Adults and Companions: Protocol for a Randomized Controlled Trial." JMIR Research Protocols, 15(1), e83864, 2026.

J3 Tauseef, M., Ullal, A., Watkins, A., et al. "Older and younger adults' perceptions of augmented reality photorealistic avatars as a viable medium for interpersonal communication." Innovation in Aging, 9(9), 2025.

J4 Watkins, A., Ghosh, R., Ullal, A., et al. "Instilling the perception of weight in augmented reality using minimal haptic feedback." Scientific Reports, 14, 24894, 2024.

J5 Watkins, A., Ullal, A., Sarkar, N. "Every "Body" Gets a Say: An Augmented Optimization Metric to Preserve Body Pose During Avatar Adaptation in Mixed/Augmented Reality." IEEE Transactions on Visualization and Computer Graphics, 31(7), 3897-3912, 2025.

J6 Ullal, A., Watkins, A., Sarkar, N. "An Efficient Redirection of Positional Interactions in Mixed/Augmented Reality." International Journal of Semantic Computing, 17(1), 95-118, 2023.

J7 Migovich, M., Ullal, A., Fu, C. "Feasibility of wearable devices and machine learning for sleep classification in children with Rett syndrome: A pilot study." Digital Health, 2023.

J8 Fan, J., Ullal, A., Beuscher, L., et al. "Field testing of Ro-Tri, a robot-mediated triadic interaction for older adults." International Journal of Social Robotics, 13, 1711-1727, 2021.

J9 Fan, J., Mion, L. C., Beuscher, L., Ullal, A., et al. "SAR-connect: A socially assistive robotic system to support activity and social engagement of older adults." IEEE Transactions on Robotics, 38(2), 1250-1269, 2021.

J10 Nie, G., Ullal, A., et al. "An Immersive Computer-Mediated Caregiver-Child Interaction System for Young Children with Autism Spectrum Disorder." IEEE Transactions on Neural Systems and Rehabilitation Engineering, 29, 884-893, 2021.

J11 Ullal, A., Su, B. Y., et al. "Non-invasive monitoring of vital signs for older adults using recliner chairs." Health and Technology, 11(1), 169-184, 2021.

Manuscripts Under Review / Revision

R1 Sarkar, N., Tate, J. A., Tauseef, M., Ullal, A., Maxwell, C. A., Juckett, L. A., Scharre, D. W., Tan, A., Ingram, Z., Zhang, R., Mion, L. C. "Feasibility and Acceptability of Collaborative Augmented Reality for Virtual Social Visits between Older Adults and Family or Friends: A Pilot Study." Geriatric Nursing (major revision resubmitted; Manuscript No. GN-D-26-00506R1).

Peer-Reviewed Conference Proceedings

C1 Tauseef, M., Watkins, A., Ingram, Z., Zhang, R., Ullal, A., Ghosh, R., Weitlauf, A. S., et al. "Autistic Adults' Perspectives on Self-Expression and Dyadic vs. Group Interactions in AR Telepresence." Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems, 2026.

C2 Ullal, A., Tauseef, M., Watkins, A., et al. "An Iterative Participatory Design Approach to Develop Collaborative Augmented Reality Activities for Older Adults in Long-Term Care Facilities." Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems, 1-21, 2024.

C3 Ullal, A., Watkins, A., Sarkar, N. "A Multi-Objective Optimization Framework for Redirecting Pointing Gestures in Remote-Local Mixed/Augmented Reality." ACM Symposium on Spatial User Interaction (SUI), 1-11, 2022.

C4 Ullal, A., Watkins, C., Sarkar, N. "A Dynamically Weighted Multi-Objective Optimization Approach to Positional Interactions in Remote-Local Augmented/Mixed Reality." IEEE AIVR, 29-37, 2021.

C5 Ullal, A., et al. "Design of an intelligent and immersive system to facilitate the social interaction between caregivers and young children with autism." HCI International, 123-132, 2019.

Book Chapter

B1 Ullal, A., Pachori, R. "Variational mode decomposition based automated diagnosis method for epilepsy using EEG signals." In Deep Learning, Machine Learning and IoT in Biomedical and Health Informatics, CRC Press, 89-111, 2022.

Peer-Reviewed Abstracts / Extended Abstracts

A1 Tauseef, M., Ullal, A., Watkins, A., et al. "From Lab to a Long-Term Care Facility: Lessons Learned from Field Deployment of Augmented Reality Telepresence System as an Interactive Communication Technology." IEEE ISMAR-Adjunct, 505-508, 2024.

A2 Maxwell, C., Tauseef, M., Ullal, A., et al. "Designing Collaborative Augmented Reality Activities with Older Adults in Long-Term Care." Innovation in Aging, 7, 1017, 2023.

A3 Ullal, A., Sarkar, N. "A Piecewise Approach to Mapping Interactions between Room-scale Environments in Remote Mixed/Augmented Reality." IEEE VRW, 663-664, 2023.

A4 Ullal, A., Sarkar, N. "Mapping of Locomotion Paths between Remote Environments in Mixed Reality using Mesh Deformation." ACM VRST, 1-2, 2022.

ACADEMIC SERVICE

Editorial and Conference Service

- Topic Editor, Frontiers in Computer Science - Human-Media Interaction, Research Topic: "Human-Centered Vision-Language-Action Models for Embodied Interaction in Augmented, Virtual, Mixed, and Smart Environments," 2026-Present.
- Guest Editor, Frontiers in Aging, 2026-Present - conducting and guiding manuscript peer review.
- Guest Editor, Frontiers in Virtual Reality, 2026-Present - conducting and guiding manuscript peer review.
- Associate Chair, ACM CHI Conference on Human Factors in Computing Systems (CHI 2027), 2026-2027 - managing external reviews, discussion, meta-review, and recommendation for assigned submissions.
- Track 2 Co-Chair (Work-in-Progress / Demos), IEEE International Conference on Artificial Intelligence and Virtual Reality (AIVR 2027), 2026-2027.

Peer Review Service

- Completed 72 peer reviews through October 2026: 38 conference reviews and 34 journal reviews (documented review log available).
- Conference reviewing: ACM CHI (2024-2026); IEEE VR (2023, 2024, 2027); ACM/IEEE HRI (2025-2026); ICWSM (2023); ACM CSCW (2025); ACM SUI (2025-2026); ACM IUI (2025-2026); IEEE RO-MAN (2025); ACM VRST (2026).
- Journal reviewing includes Scientific Reports; IEEE Transactions on Computational Social Systems; Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT); International Journal of Social Robotics; International Journal of Human-Computer Interaction; Universal Access in the Information Society; Displays; Informatica; Behavioral Sciences; Journal of Eye Movement Research; Computers; JMIR Formative Research; BMC Geriatrics; Virtual Worlds; International Journal of Geo-Information; and multiple Frontiers journals including Computer Science, Psychology, Public Health, Robotics and AI, Virtual Reality, Sustainable Cities, Mechanical Engineering, Dementia, and Sports and Active Living.

TEACHING AND INSTRUCTION

- Instructor, AI Passport Fall 2026, University of Florida, Fall 2026.
- Instructor, CAI 6734 - Applied Generative AI in Medicine, University of Florida, Fall 2026; instruction includes medical LLMs, model adaptation/fine-tuning, reinforcement learning, agentic AI, evaluation, and AI safety/governance.
- Teaching support, Introduction to Logic Systems (2210), University of Missouri, Spring 2016.
- Teaching support, Capstone Project (ECE 3110), University of Missouri, Fall 2017.
- Teaching support, Senior Capstone Project Design (ECE 4970), University of Missouri, Spring 2017 and Spring 2018.

HONORS AND AWARDS

- Best Student Paper Award, IEEE International Conference on Artificial Intelligence and Virtual Reality (AIVR), 2021.
- Top 4 Finalist, Best Paper Award, IEEE AIVR, 2021.
- Graduate Travel Grant, Vanderbilt University, for IEEE AIVR 2021, IEEE VR 2023, and ACM CHI 2024.
- Deaton Scholar, Brady and Anne Deaton Institute.
- Curators Grant-in-Aid Scholarship, University of Missouri.

FUNDING AND GRANT CONTRIBUTIONS

- Contributed to proposal development for NSF Smart and Connected Communities grant #2225890, "Reducing Loneliness for Long Term Care Older Adults through Collaborative Augmented Reality," funded for approximately $2.25M.
- Contributed to proposal development for NIH/NIA R21 grant #1R21AG078480-01, "Reducing Loneliness of Older Adults in Long Term Care Facilities through Collaborative Augmented Reality," funded for $433,495.

MENTORSHIP

- Ph.D. student mentorship: Mahrukh Tauseef (Vanderbilt University).
- Master's student mentorship: Jiaqi Liu (Vanderbilt University); Anina Pillai (University of Florida).
- Undergraduate mentorship: Nathaniel Lathrop, Zalen Ingram, Isaac Shirk, Gwyneth Polak, Kyle Nicholson, Bryan Schuder, Narayan Rizal, Jonah Weaver, Martin Weaver (Middle Tennessee State University); Gabriel Suvadjzic (University of Florida).

INVITED TALKS

- "Use of Collaborative AR Telepresence Technologies to Reduce Loneliness in Older Adults," Rutherford County Chamber of Commerce, Murfreesboro, TN.
- "Potential of Avatar-Based Telepresence Technologies for Interventions with Autistic Populations," Multisensory Research Lab, Vanderbilt University.
- "Towards Mixed Reality as a Substitute for In-Person Interaction," The Institution of Engineers (India), Mangalore, India.

TECHNICAL SKILLS

Programming: Python, C/C++, C#, MATLAB, SQL, Java, JavaScript/HTML/CSS.
AI / Machine Learning: PyTorch, TensorFlow, Transformers, large language models, retrieval-augmented generation, multimodal learning, supervised/unsupervised learning, optimization, reinforcement learning.
XR / Embodied Systems: Unreal Engine, Unity, NVIDIA Omniverse/OpenUSD, MetaHuman, AR/VR interaction, telepresence, avatars, 3D user interfaces.
Clinical / Data Systems: Electronic health records, longitudinal multimodal data, real-time inference pipelines, HL7-oriented clinical data integration.
Sensing / Modeling: EEG, PPG, IMUs, actigraphy, biomedical signal processing; SOLIDWORKS, COMSOL Multiphysics, Blender, Maya, Adobe Photoshop.